

VISVESVARAYA TECHNOLOGICAL UNIVERSITY

Jnana Sangama, Belgaum -590014, Karnataka State, India



A PROJECT REPORT ON

“G-care: An AI Healthcare Assistant”

BACHELOR OF ENGINEERING

IN

DEPARTMENT

OF

COMPUTER SCIENCE ENGINEERING [AI&ML]

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AY 2026-27

CERTIFICATE

This is to certify that the project report entitled G-care: An AI Healthcare Assistant submitted by Yeshrutha S and Abhay Vadagi bearing the USN 1EW23CI063 and 1EW23CI002 to the Visvesvaraya Technological University, Belagavi, in partial fulfillment for the award of the degree of B. Tech in CSE[AI&ML] is a *bona fide* record of project work carried out by her under my supervision. The contents of this report, in full or in parts, have not been submitted to any other Institution or University for the award of any degree.

Signature:

Name of the Guide: Asst. Prof. Sreehari M

Department: CSE[AI&ML]

Signature of HOD with seal

DECLARATION

I/We declare that this project report titled G-care: An AI Healthcare Assistant submitted in partial fulfillment of the degree of B. Tech in CSE[AI&ML] is a record of original work carried out by me under the supervision of Sreehari M, Assitant Professor, CSE[AI&ML], and has not formed the basis for the award of any other degree, in this or any other Institution or University. In keeping with the ethical practice in reporting scientific information, due acknowledgements have been made wherever the findings of others have been cited.

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CERTIFICATE

Certified that the project work entitled G-care: An AI Healthcare Assistant was carried out by Mr./Ms. Yeshrutha S and Abhay Vadagi, USN 1EW23CI063 and 1EW23CI063, a bonafide student of Asst. Prof. Sreehari M in partial fulfilment for the award of Bachelor of Engineering / Bachelor of Technology in CSE[AI&ML] of the Visveswaraiah Technological University, Belgaum during the year 2026-27. It is certified that all corrections/suggestions indicated for the Internal Assessment have been incorporated in the Report deposited in the departmental library. The project report has been approved as it satisfies the academic requirements in respect of Project work prescribed for the said Degree.

Name & Signature of the Guide

Name Signature of the HOD

Signature of the Principal

External Viva

Sl. No. Name of the examiners

Signature with date

1

2

ABSTRACT

With the rising global geriatric population and the prevalence of nuclear family structures, ensuring the continuous health monitoring, safety, and independence of senior citizens has emerged as a critical societal and technological challenge. Elderly individuals living independently frequently face life-threatening risks due to unmonitored chronic conditions, missed medication schedules, undetected physical falls, and delayed emergency responses. Existing remote healthcare systems often suffer from high operational costs, steep learning curves, fragmented communication, and interfaces unsuitable for seniors.

To resolve these limitations, this project presents **G-Care (Guardian Care)**, an integrated, full-stack, and intelligent remote elderly care platform that bridges the communication gap between elderly patients, family guardians, and healthcare practitioners. The system incorporates a tri-tier, Role-Based Access Control (RBAC) architecture featuring a **Senior Wearable Interface / Simulator**, a **Caretaker & Guardian Dashboard**, and a **Specialized Clinical Doctor Portal**. Real-time physiological telemetry—including heart rate, peripheral oxygen saturation (SpO₂), blood pressure, body temperature, stress, and hydration—is monitored continuously and processed through real-time anomaly detection heuristics to immediately identify critical vitals deviations, physical falls, and geofence breaches.

The platform provides an automated medication and routine scheduling engine with interactive adherence check-offs, instant SOS emergency dispatch protocols, and senior-centric accessibility features including a multilingual voice assistant supporting regional Indian languages (English, Hindi, Kannada, and Tamil). Developed using React 18, TypeScript, Tailwind CSS, Zustand, and Node.js/Express with PostgreSQL and JWT authentication, G-Care provides a scalable, secure, and cost-effective solution for proactive geriatric telecare. Experimental evaluations demonstrate responsive vitals synchronization, low alert dispatch latency, and significant reduction in caregiver dependency.

Keywords: *Elderly Care Monitoring, Telehealth, Internet of Medical Things (IoMT), Vitals Anomaly Detection, Role-Based Access Control, Medication Adherence, Fall Detection, Multilingual Voice Assistant.*

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LIST OF FIGURES

Figure No	Name	Page No
1.1	High-Level Conceptual Architecture and Dataflow of the G-Care Ecosystem	5
3.1	Multi-Tier Layered System Architecture of G-Care	11
3.2	DFD Level 0 — Context Level Diagram	14
3.3	DFD Level 1 — Detailed Functional Data Flow Diagram	15
5.1	End-to-End Operational Methodology Flowchart of G-Care	23
5.2	Shared I2C Bus Interfacing on Prototype Breadboard	25
5.3	Entity-Relationship Diagram of G-Care Platform	26
5.4	Use Case Diagram for the Tri-Tier G-Care System	27
7.1	Multi-Tier Testing Pipeline for G-Care Platform	38

LIST OF TABLES

Table No	Name	Page No
2.1	Comparison Matrix of State-of-the-Art Solutions vs. Proposed G-Care Platform	8
4.1	Edge Hardware and Prototyping Components	16
4.2	Software Technology Stack	18
6.1	Breadboard Pin Interfacing and Wiring Connections	28
7.1	Comprehensive System Test Cases and Empirical Verification Results	39

CONTENTS

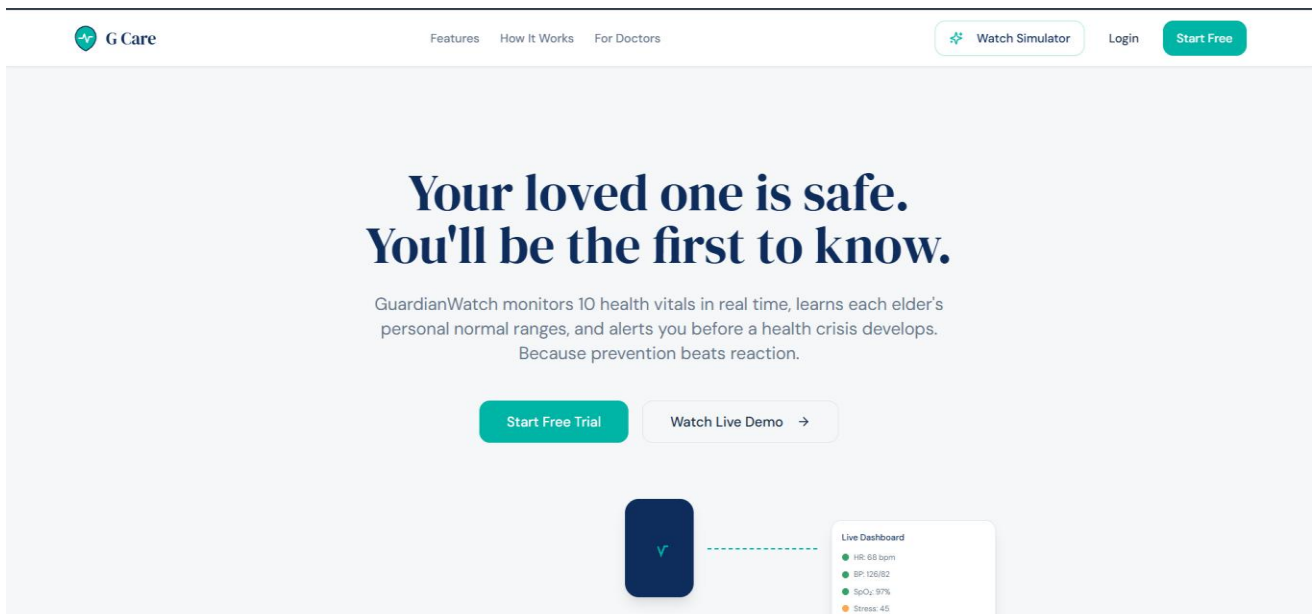
Chapter No.	Chapter Name	Page No.
CHAPTER 1	INTRODUCTION	1
	1.1	2
	1.2	3
CHAPTER 2	LITERATURE REVIEW	6
	2.1	6
	2.2	8
	2.3	9
CHAPTER 3	SYSTEM ARCHITECTURE	11
	3.1	11
	3.2	12
	3.3	14
CHAPTER 4	SYSTEM REQUIREMENTS	16
	4.1	16
	4.2	18
	4.3	20
CHAPTER 5	METHODOLOGY	22
	5.1	22
	5.2	24
CHAPTER 6	IMPLEMENTATION	28
	6.1	28
	6.2	32
	6.3	36
CHAPTER 7	TESTING	38
	7.1	38
	7.2	39
CHAPTER 8	CONCLUSION AND FUTURE SCOPE	42
	BIBILIOGRAPY	45
	APPENDIX I	46

BIBLIOGRAPHY

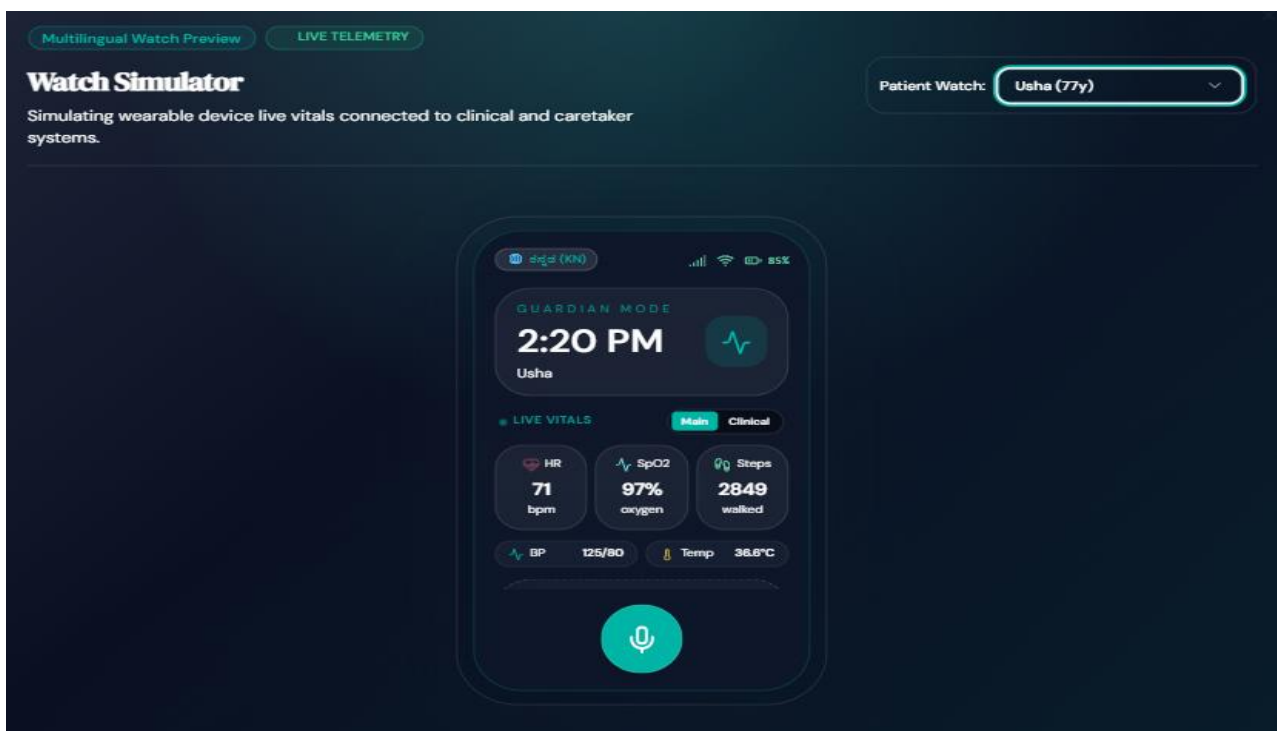
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APPENDIX A

SNAPSHOTS



G Care Telecare Platform



G-Care Multilingual Watch Simulator Interface


G Care

Every second matters.

- 10 vitals tracked
- 4 languages
- < 3s alerts
- Guardian Portal

Caretaker
Doctor
Guardian

Welcome back

Email address

Password

☐ Remember me [Forgot password?](#)

Log In

New to GuardianWatch? [Create an account](#)

Multi-Role User Authentication and Login Interface


G Care

Yeshrutha S
Caretaker

- Dashboard
- Elders
- Medications
- Alarms
- Alerts 99
- Reports
- Doctor Portal
- Settings

Dashboard

AI risk index

24

Weighted from vitals, battery, alerts, and baseline status.

Live devices

3/3

Connected watches streaming recent health signals.

Unresolved alerts

99

Active care events that still need attention.

Stable profiles

2

0 watch needs charging soon.

Highest priority

Venkatesh Rao

Check breathing comfort and keep oxygen trend under review.

SpO2
92.6%

BP
120/77

Stress
61/100

Review vitals
Call family
Locate

Watch 57

AI care queue

Ranked by current risk and monitoring reliability.

VR
Venkatesh Rao
Age 71

HR 81
SPO₂ 92.6%
91%

Check breathing comfort and keep oxygen trend under...
Risk: 57

U
Usha
Age 77

HR 72
SPO₂ 97%
78%

Vitals are inside personal range. Keep routine monitoring active.
Risk: 8

LD
Lakshmi Devi
Age 82

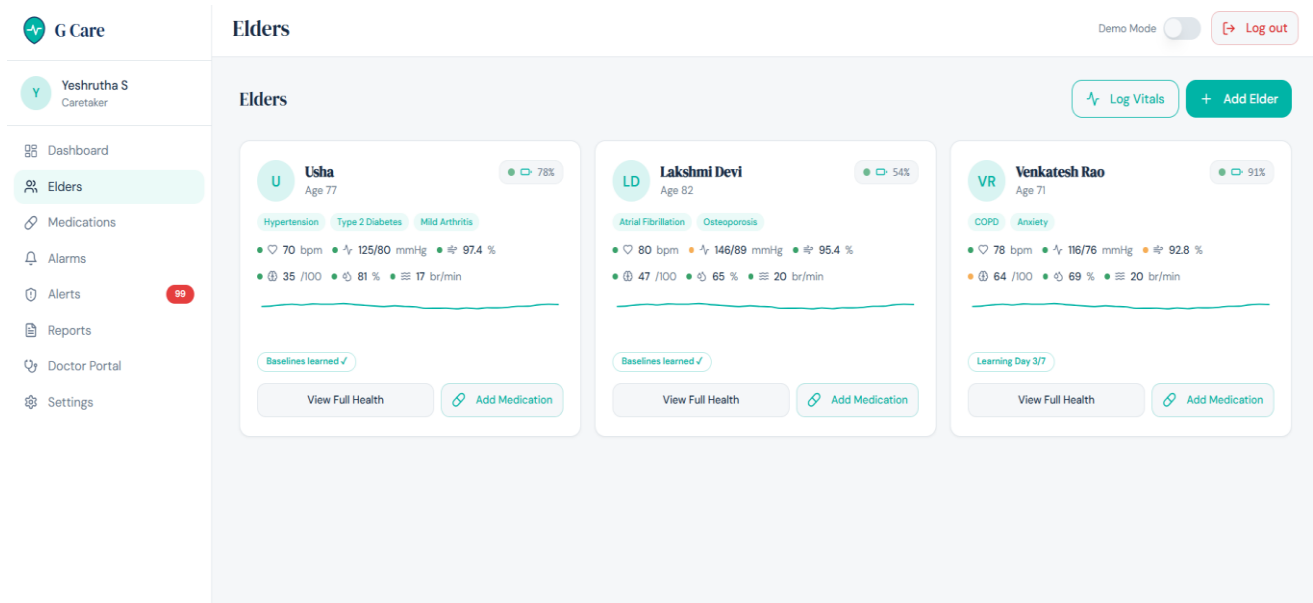
HR 79
SPO₂ 97.1%
54%

Vitals are inside personal range. Keep routine monitorin...
Risk: 8

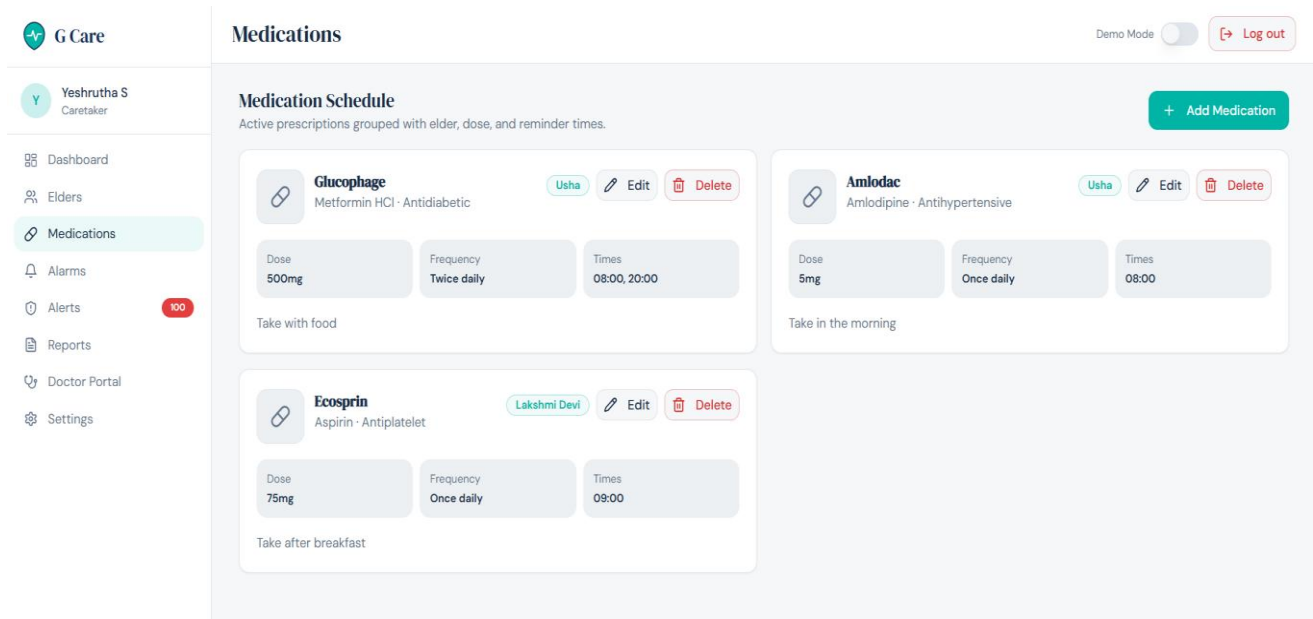
Elders

Log Vitals
Add Elder

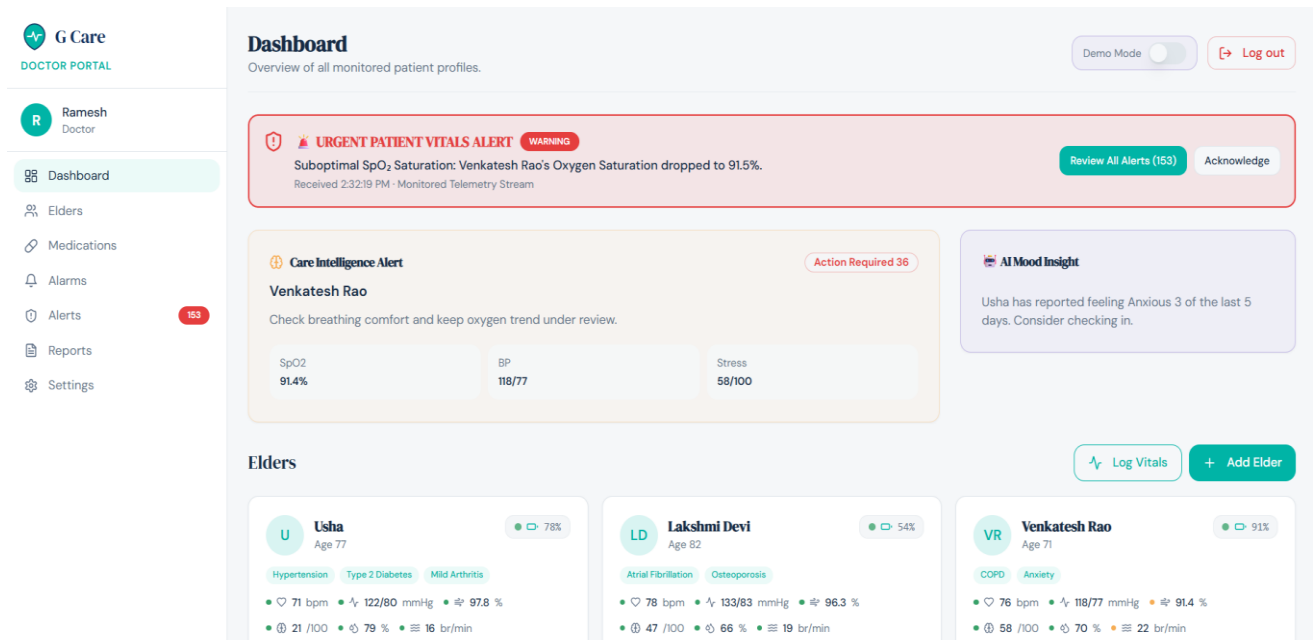
Caretaker Dashboard for Elder Health Monitoring



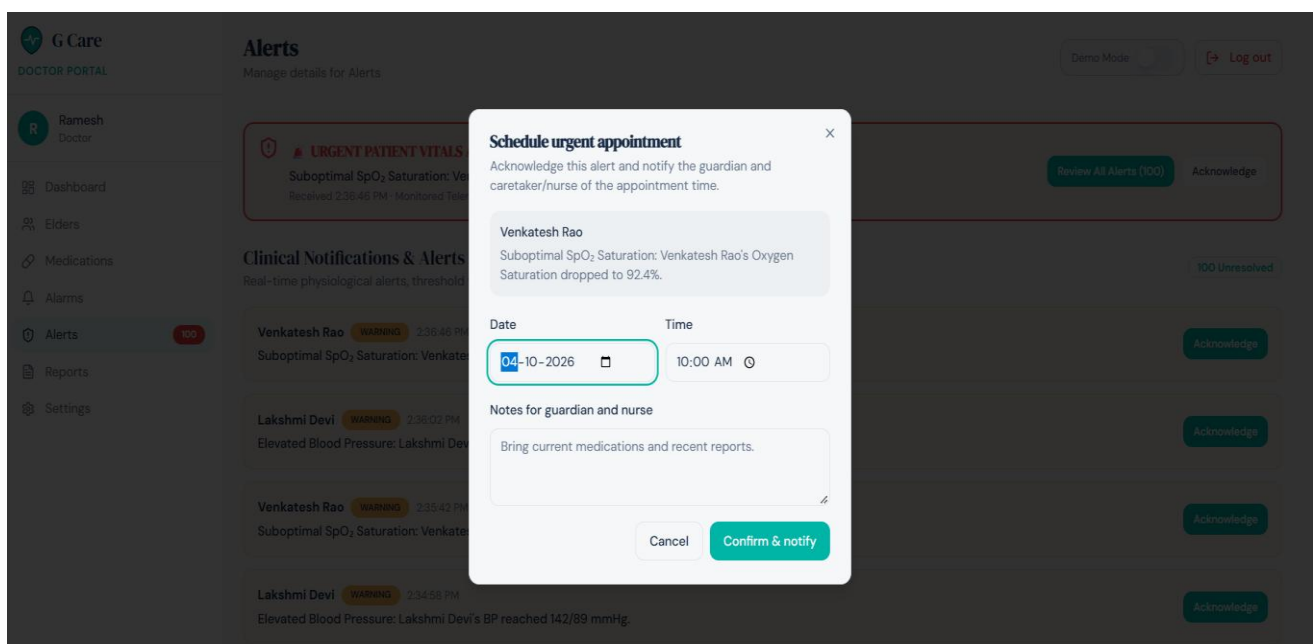
Caretaker Elders Health Monitoring Dashboard



Caretaker Medication Schedule and Management Interface



Doctor Portal Dashboard for Patient Monitoring and Clinical Alerts



Clinical Alert Management and Urgent Appointment Scheduling

**VITAL ANOMALY ALERT**

WARNING

Usha: Vital Alert: Usha's biometric stress index spiked to 90/100. Dr. Ramesh Kumar and caretaker have been notified.

Logged at 2:23:50 PM · Clinical Doctor & Caretaker Notified

Acknowledge & Confirm Safe

View Alerts (12)

Feed

Logs

Alerts

Reminders

SOS

Smart TV

● Watch Connected — Live

Motion Monitor

Sitting

Accelerometer + Gyroscope

Shiver Monitor

Normal

Micro-tremor analysis

Heart Rate

71 bpm

Blood Pressure

128/83 mmHg

SpO₂

97.8 %

Stress

18 /100

Hydration

64 %

Temperature

36.6 °C

Guardian Portal for Real-Time Elder Health Monitoring